



THE KING'S SCHOOL

2010 Higher School Certificate Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Answer each question in a separate booklet

Total marks – 120

- Attempt Questions 1-8
- All questions are of equal value

Disclaimer

This is a Trial Higher School Certificate Examination only. Whilst it reflects and mirrors both the format and topics of the Higher School Certificate Examination designed by the NSW Board of Studies for the respective sections, there is no guarantee that the content of this examination exactly replicates the actual Higher School Certificate Examination.

Total marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) (i) Express $\frac{x-1}{(x+1)(x^2+1)}$ in partial fractions. 3

(ii) Find $\int \frac{x-1}{(x+1)(x^2+1)} dx$ 2

(b) (i) Show that $\cos 2\theta = \frac{1-t^2}{1+t^2}$ where $t = \tan\theta$ 2

(ii) Evaluate $\int_0^{\frac{\pi}{4}} \frac{4}{5-3\cos 2\theta} d\theta$ 3

(c) Evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin 2\theta}{\sqrt{1-\sin^4\theta}} d\theta$ 3

(d) By using the result $a - \frac{a}{a+1} = \frac{a^2}{a+1}$, or otherwise, find $\int \frac{e^{-2x}}{e^{-x}+1} dx$ 2

End of Question 1

(a) Let $z = 3 - 2i$ and $w = -1 + 3i$

Find

(i) $\overline{z + w}$ **1**

(ii) $|zw|$ **2**

(b) Let $z = 1 + ai$ and $w = a + i$, a real

Find $\arg(zw)$ **2**

(c) (i) Expand $(a + ib)^3$ **1**

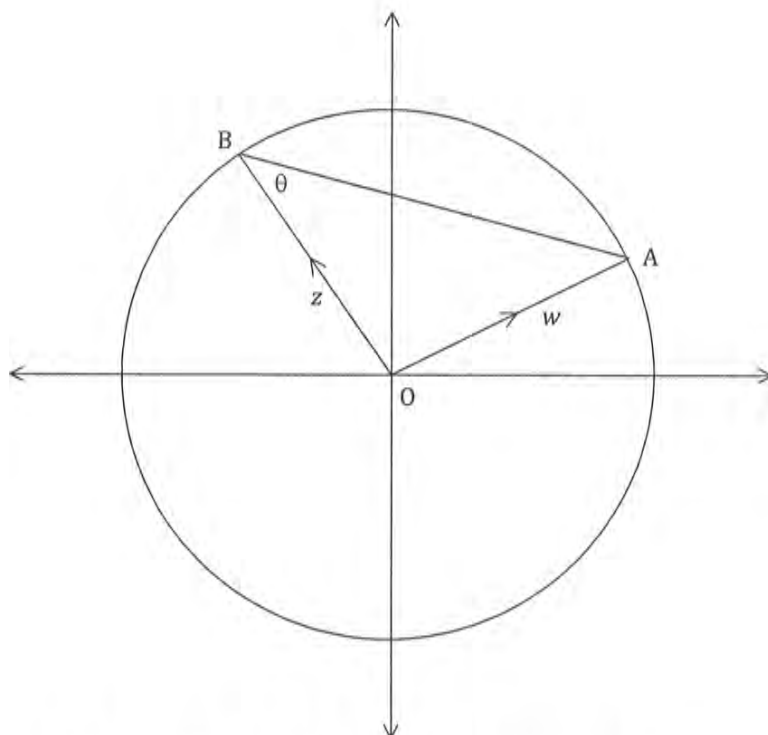
(ii) If $(a + ib)^3 = x + iy$, a, b, x, y real,

show that $\frac{x}{a} + \frac{y}{b} = 4(a^2 - b^2)$, $a, b \neq 0$ **2**

(d) Sketch the region in the complex plane where $|z - 2i| \geq |z + 2|$ **2**

Question 2 continues on the next page

(e)



The diagram shows two distinct points A and B that represent the complex numbers w and z respectively. These points lie on the circle with centre O and radius r .

Let $\angle OBA = \theta$

(i) Show that $z = w(-\cos 2\theta + i\sin 2\theta)$ 2

(ii) Write down the complex number represented by $\vec{AB}$ and deduce that $|z + w| = \tan\theta |z - w|$ 3

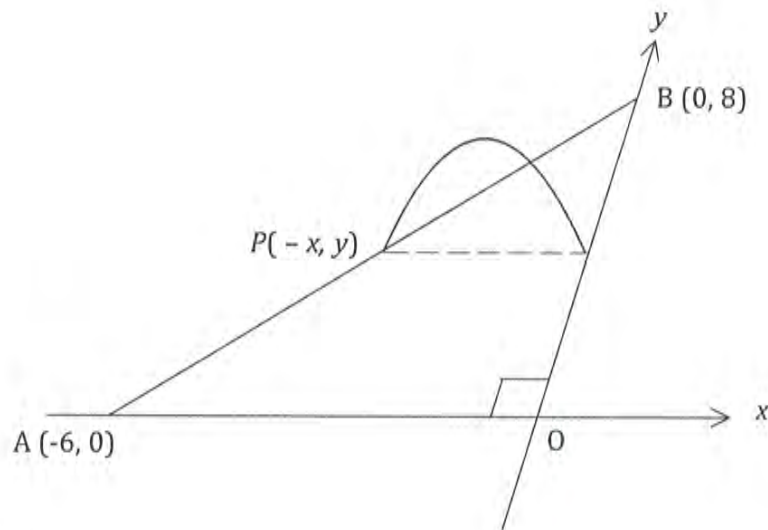
End of Question 2

- (a) Find the gradient of the curve defined by $(x + 1)y + \cos \pi y + x^3 = 0$
at the point where $y = 1$ **4**
- (b) Let $f(x) = \ln(\sqrt{x^2 + 1} - x)$
- (i) Show that x can assume any real value. **1**
- (ii) Show that the function is odd. **2**
- (iii) Show that $f'(x) = -\frac{1}{\sqrt{x^2 + 1}}$ **2**
- (iv) Given that $f''(x) = \frac{x}{(x^2 + 1)^{\frac{3}{2}}}$,
sketch the graph of $y = f(x)$ **2**
- (v) Use integration by parts to evaluate $\int_0^1 \ln(\sqrt{x^2 + 1} - x) dx$ **4**

End of Question 3

- (a) Prove that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$, is πab 3

(b)



The base of a solid is the triangular region AOB where $A = (-6, 0)$, $O = (0, 0)$ and $B = (0, 8)$.

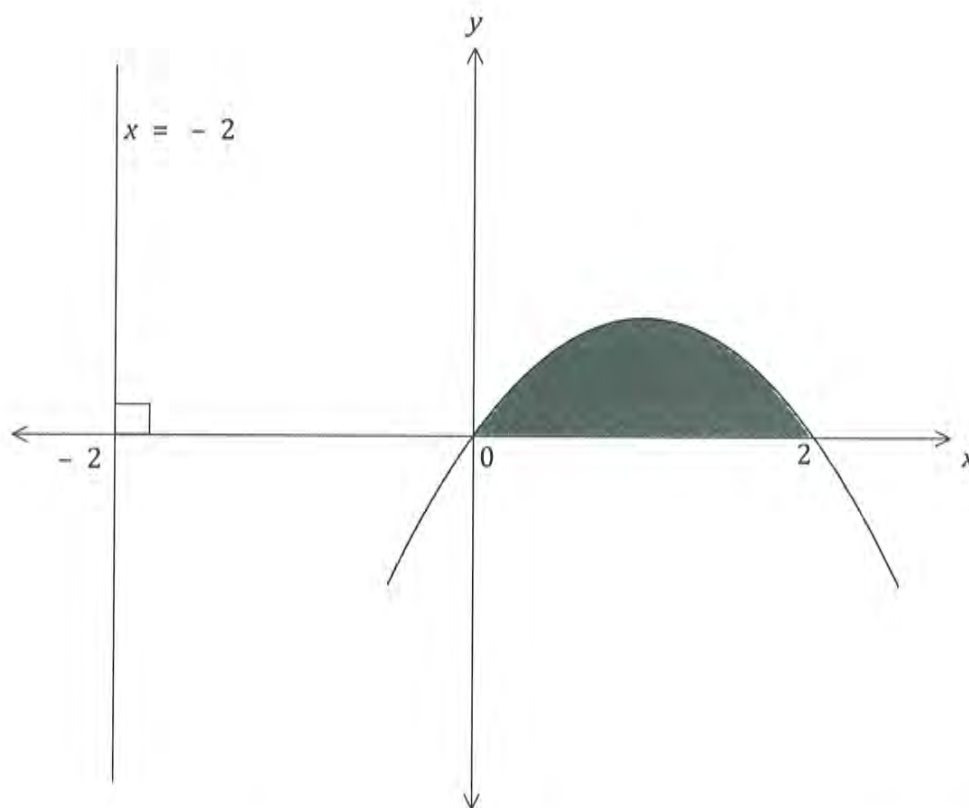
Vertical cross-sections of the solid parallel to AO are semi-ellipses where the major axis lies in the base of the solid and is twice the length of the minor axis.

Let $P(-x, y)$ be a point on AB. Find the volume of the solid.

5

Question 4 continues on the next page

(c)



The diagram shows the region bounded by $y = x(2 - x)$ and the x axis, $0 \leq x \leq 2$

(i) Without using integration, show that the area of the region is $\frac{4}{3}$ 2

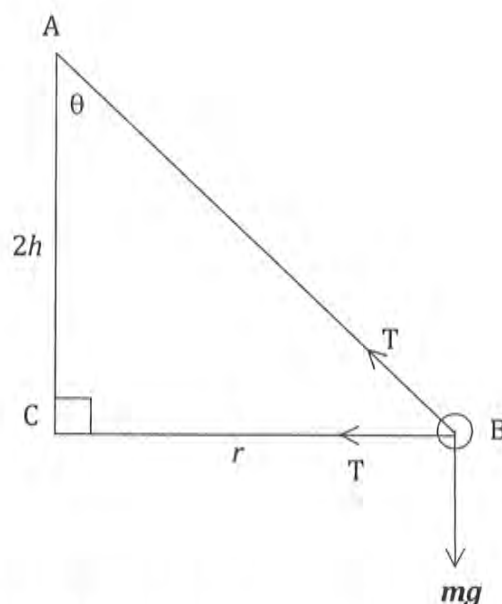
(ii) The region is revolved about the line $x = -2$.

By taking slices parallel to the line $x = -2$, find the volume of the solid generated.

5

End of Question 4

(a)



A string ABC of length $2l$ is attached at two points A and C. C is $2h$ vertically below A.

A smooth ring of mass m is free to move on the string and rotates about C in a horizontal plane with constant angular velocity ω .

Let $\angle CAB = \theta$ and the radius of revolution CB be r .

(i) Show that $r < l$ 1

(ii) Show that $T = mg \sec \theta$ 1

(iii) Show that $\omega^2 r = g(\sec \theta + \tan \theta)$ 2

(iv) Prove that $r = \frac{gl}{h\omega^2}$ 2

(v) Deduce that $\omega > \sqrt{\frac{g}{h}}$ 1

Question 5 continues on the next page

- (b) (i) Prove by induction for positive integers n that

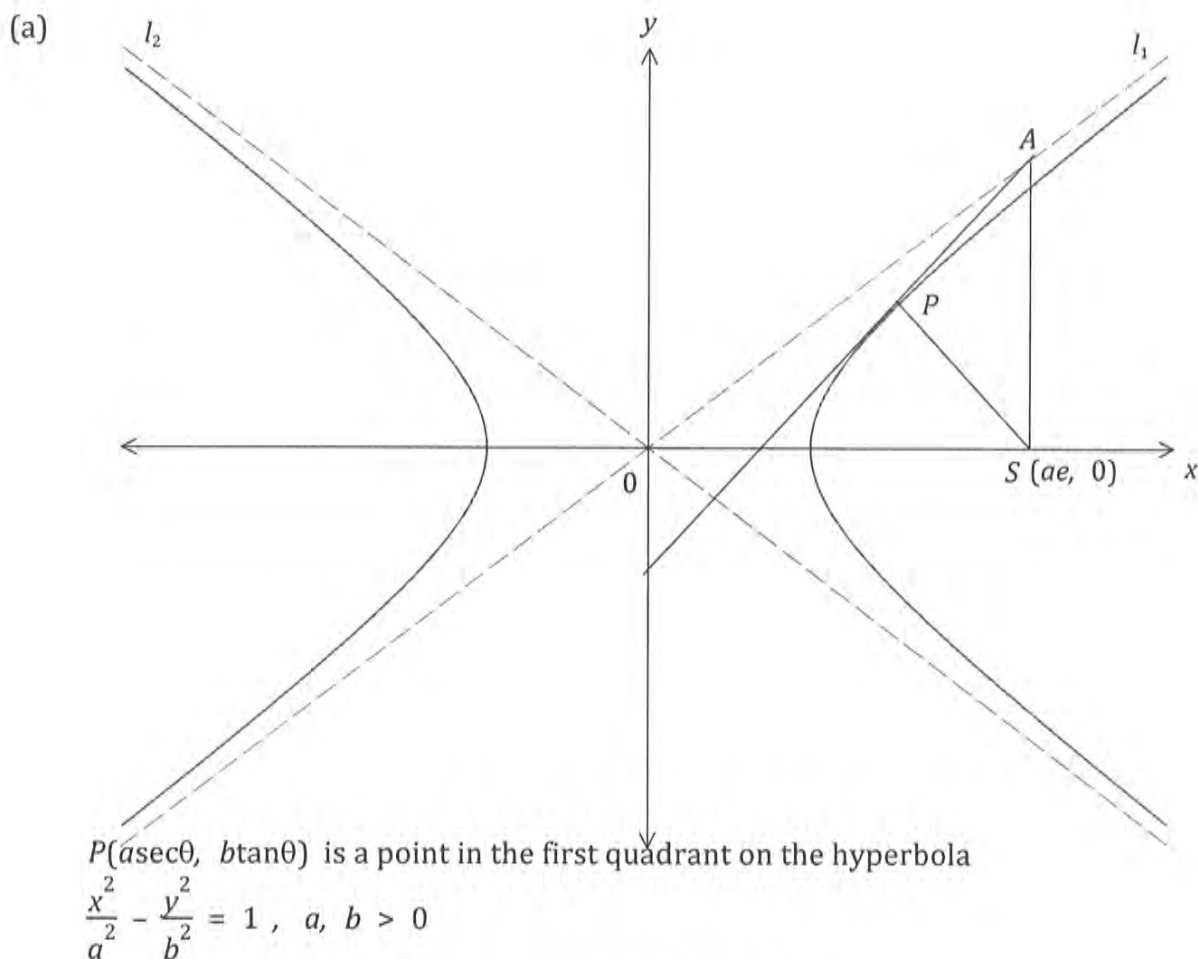
$$\frac{4.1}{1.3} - \frac{4.2}{3.5} + \frac{4.3}{5.7} - \dots + \frac{(-1)^n 4(n+1)}{(2n+1)(2n+3)} = 1 + (-1)^n \frac{1}{2n+3} \quad 4$$

- (ii) Find the sum $\frac{4.1}{1.3} - \frac{4.2}{3.5} + \frac{4.3}{5.7} - \dots$ 1

- (c) Let $z = \cos\theta + i\sin\theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

Find the modulus and argument of the complex number $z^2 + 1$ 3

End of Question 5



$S(ae, 0)$ is a focus of the hyperbola, where e is the eccentricity.

l_1, l_2 are the asymptotes of the hyperbola.

The line $x = ae$ and the tangent at P meet at A on l_1

- | | |
|--|---|
| (i) Show that $A = (ae, be)$ | 2 |
| (ii) Prove that the equation of the tangent at $P(a \sec \theta, b \tan \theta)$ is
$\frac{\sec \theta}{a} x - \frac{\tan \theta}{b} y = \sec^2 \theta - \tan^2 \theta$ | 2 |
| (iii) Prove that $\sec \theta + \tan \theta = e$ | 2 |
| (iv) Deduce that SP is parallel to l_2 | 1 |

Question 6 continues next page

- (b) A particle moves on the x axis so that at anytime t its acceleration $\ddot{x}$ is given by $\ddot{x} = \frac{16}{v} + v$, where v is the velocity.

Initially, $v = V > 0$

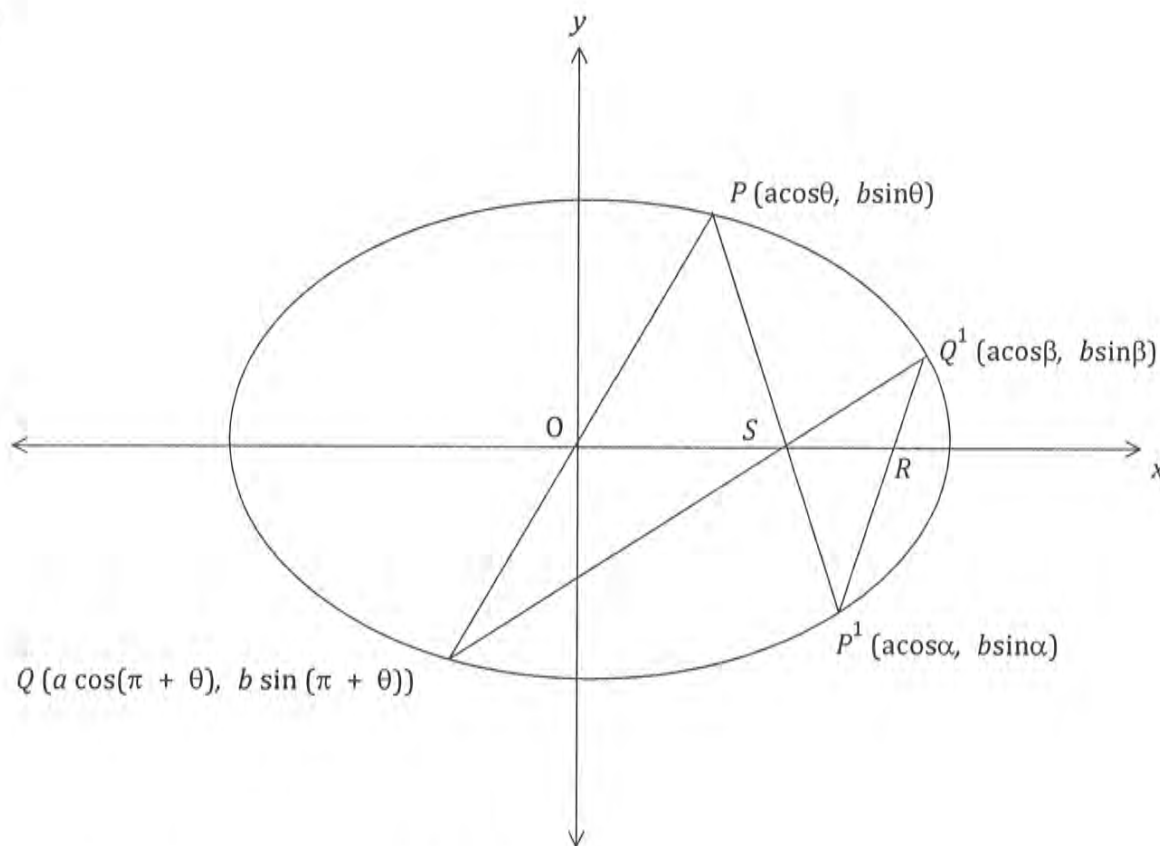
Suppose the particle travels a distance X in time T to acquire a velocity of $4V$.

- (i) Prove that $T < 2 \ln 2$ 4

- (ii) Prove that $X = 3V - 4 \tan^{-1} \left(\frac{3V}{4 + V^2} \right)$ 4

End of Question 6

(a)



PQ is a chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$, passing through $O(0, 0)$

PP^1 and QQ^1 are focal chords passing through the focus $S(ae, 0)$, e the eccentricity.

The equation of the chord P^1Q^1 is

$$\frac{x}{a} \cos \left(\frac{\alpha + \beta}{2} \right) + \frac{y}{b} \sin \left(\frac{\alpha + \beta}{2} \right) = \cos \left(\frac{\alpha - \beta}{2} \right)$$

[DO NOT PROVE THIS]

The chord P^1Q^1 meets the major axis at R .

Question 7 continues on the next page

- (a) (i) Show that at R the x coordinate is given by

$$x = a \left(\frac{1 + \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}} \right) \quad 2$$

- (ii) Use the focal chord PP' to show that

$$e \cos \left(\frac{\alpha + \theta}{2} \right) = \cos \left(\frac{\alpha - \theta}{2} \right) \quad 1$$

(iii) Deduce that $\tan \frac{\alpha}{2} \tan \frac{\theta}{2} = \frac{e - 1}{e + 1}$ 2

(iv) Hence show that $-\tan \frac{\beta}{2} \cot \frac{\theta}{2} = \frac{e - 1}{e + 1}$ 2

(v) Deduce that R is a fixed point. 2

(b) (i) Show that $(1 - x^b)^{n-1} - (1 - x^b)^n = x^b (1 - x^b)^{n-1}$ 1

Let $u_n = \int_0^1 x^{a-1} (1 - x^b)^n dx$, $n = 0, 1, 2, 3, \dots$ where a and b are positive integers.

(ii) Find u_0 1

(iii) Prove that $(a + bn)u_n = bn u_{n-1}$, $n \geq 1$ 4

End of Question 7

(a) $P(x) = 2x^3 - Ax - 2 = 0$ has the roots α, β, γ

(i) Show that $\alpha^2 + \beta^2 + \gamma^2 = A$ **2**

(ii) Show that $\frac{\beta}{\gamma} + \frac{\gamma}{\beta} = A\alpha - \alpha^3$ **2**

(iii) Find a polynomial with the three roots

$$\frac{\beta}{\gamma} + \frac{\gamma}{\beta}, \frac{\gamma}{\alpha} + \frac{\alpha}{\gamma}, \frac{\alpha}{\beta} + \frac{\beta}{\alpha}$$
4

[YOU MAY LEAVE YOUR ANSWER IN UNEXPANDED FORM]

Question 8 continues on the next page

- (b) (i) Find the sum S of the geometric series

$$1 + (1 + x) + (1 + x)^2 + \dots + (1 + x)^{n-1}, \quad x \neq 0 \quad 1$$

- (ii) Deduce that $S = \binom{n}{1} + \binom{n}{2}x + \binom{n}{3}x^2 + \dots + \binom{n}{n}x^{n-1}$ 2

- (iii) Show that

$$\begin{aligned} & \binom{n}{1} + \binom{n}{2}x + \binom{n}{3}x^2 + \dots + \binom{n}{n}x^{n-1} \\ &= 1 + (1 + x) + (1 + x)^2 + \dots + (1 + x)^{n-1} \quad \text{if } x = 0 \end{aligned} \quad 1$$

- (iv) Prove that

$$\binom{n}{1} - \frac{1}{2}\binom{n}{2} + \frac{1}{3}\binom{n}{3} - \dots + (-1)^{n+1}\frac{1}{n}\binom{n}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \quad 3$$

End of Examination Paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$



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Mathematics Extension 2

Question (Marks)	Complex Numbers	Functions	Integration	Conics	Mechanics
1 (15)			(a)-(d) 15		
2 (15)	(a)-(d) 15				
3 (15)		(a),(b)(i)-(iv) 11	(b)(v) 4		
4 (15)			(a)-(c) 15		
5 (15)	(c) 3	(b) 5			(a) 7
6 (15)				(a) 7	(b) 8
7 (15)			(b) 6	(a) 9	
8 (15)		(a), (b) 15			
Total (120)	18	31	40	16	15

Question 1

(a) (i) Put $\frac{x-1}{(x+1)(x^2+1)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$

$$\therefore A(x^2+1) + (Bx+C)(x+1) \equiv x-1$$

$$x = -1, \quad 2A = -2, \quad A = -1$$

$$\therefore -1 + B = 0, \quad B = 1$$

$$\therefore -1 + C = -1, \quad C = 0$$

$$\Rightarrow \frac{-1}{x+1} + \frac{x}{x^2+1}$$

$$(ii) \quad I = \int \frac{x}{x^2+1} - \frac{1}{x+1} dx = \frac{1}{2} \ln(x^2+1) - \ln(x+1) + C$$

(b) (i) $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = \frac{1 - \tan^2 \theta}{\sec^2 \theta} = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1 - t^2}{1 + t^2}$

(ii) Put $t = \tan \theta$

$$\theta = 0, \quad t = 0$$

$$\frac{dt}{d\theta} = \sec^2 \theta = 1 + t^2$$

$$\theta = \frac{\pi}{4}, \quad t = 1$$

$$\therefore I = \int_0^1 \frac{4}{5 - 3 \cdot \frac{1-t^2}{1+t^2}} \frac{dt}{1+t^2}$$

$$= \int_0^1 \frac{4}{5 + 5t^2 - 3 + 3t^2} dt$$

$$= \int_0^1 \frac{2}{1 + 4t^2} dt$$

$$= 2 \cdot \frac{1}{2} [\tan^{-1} 2t]_0^1 = \tan^{-1} 2$$

(c) Put $u = \sin^2 \theta$

$\theta = 0, u = 0$

$$\frac{du}{d\theta} = 2 \sin \theta \cos \theta = \sin 2\theta$$

$\theta = \frac{\pi}{4}, u = \frac{1}{2}$

$$\therefore I = \int_0^{\frac{1}{2}} \frac{du}{\sqrt{1-u^2}} = \left[\sin^{-1} u \right]_0^{\frac{1}{2}} = \frac{\pi}{6}$$

(d) $I = \int e^{-x} - \frac{e^{-x}}{e^{-x} + 1} dx$

$$= -e^{-x} + \ln(e^{-x} + 1) + C$$

Question 2

(a) (i) $z + w = 2 + i \quad \therefore \overline{z + w} = 2 - i$

(ii) $|zw| = |z||w| = \sqrt{13} \sqrt{10} = \sqrt{130}$

(b) $zw = a + i + a^2 i - a = (a^2 + 1)i$

$$\Rightarrow \arg zw = \frac{\pi}{2} \quad \text{since } a^2 + 1 > 0$$

(c) (i) $a^3 + 3a^2 ib + 3ai^2 b^2 + i^3 b^3$ will do

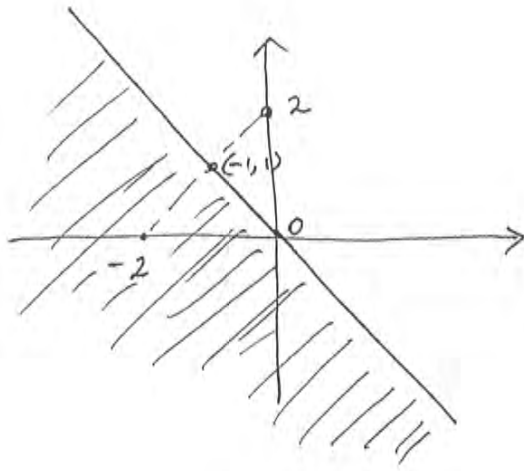
$$= a^3 - 3ab^2 + i(3a^2 b - b^3)$$

(ii) From (i) $x = a^3 - 3ab^2$

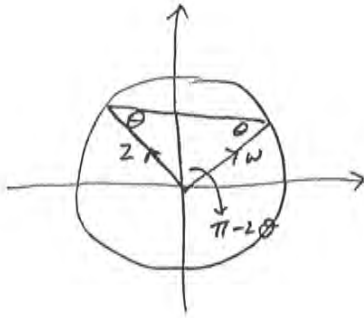
$$y = 3a^2 b - b^3$$

$$\therefore \frac{x}{a} + \frac{y}{b} = a^2 - 3b^2 + 3a^2 - b^2 = 4(a^2 - b^2)$$

(d)



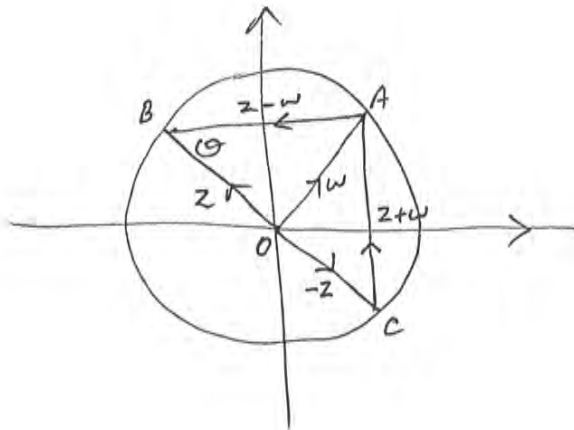
(e) (i)



$$\therefore z = w (\cos(\pi - 2\theta) + i \sin(\pi - 2\theta))$$

$$\text{i.e. } z = w (-\cos 2\theta + i \sin 2\theta)$$

(ii) $\vec{AB} = z - w$



$$\vec{OC} = -z$$

$$\vec{CA} = z + w$$

$$\angle BAC = \frac{\pi}{2} \quad (\angle \text{ in semi-circle})$$

$$\therefore \tan \theta = \frac{|z + w|}{|z - w|}$$

$$\text{i.e. } |z + w| = \tan \theta |z - w|$$

Question 3

$$(a) \quad y=1 \Rightarrow x+1 -1 + x^3 = 0$$

$$\text{ie } x(1+x^2) = 0 \Rightarrow x=0$$

$$(x+1) \frac{dy}{dx} + y - \pi \sin \pi y \frac{dy}{dx} + 3x^2 = 0$$

$$\therefore \text{ at } (0, 1), \quad \frac{dy}{dx} + 1 - 0 + 0 = 0$$

$$\frac{dy}{dx} = -1$$

$$(b) \quad (i) \quad \sqrt{x^2+1} > \sqrt{x^2}-x \geq 0 \quad \forall x \quad [\text{ie. } |x|-x \geq 0]$$

$$\Rightarrow \sqrt{x^2+1} - x > 0 \quad \forall x$$

$\therefore$ domain is all real values for x

$$(ii) \quad f(x) + f(-x) = \ln(\sqrt{x^2+1}-x) + \ln(\sqrt{x^2+1}+x)$$

$$= \ln(x^2+1-x^2)$$

$$= \ln 1 = 0$$

$\Rightarrow f(x)$ is an odd function

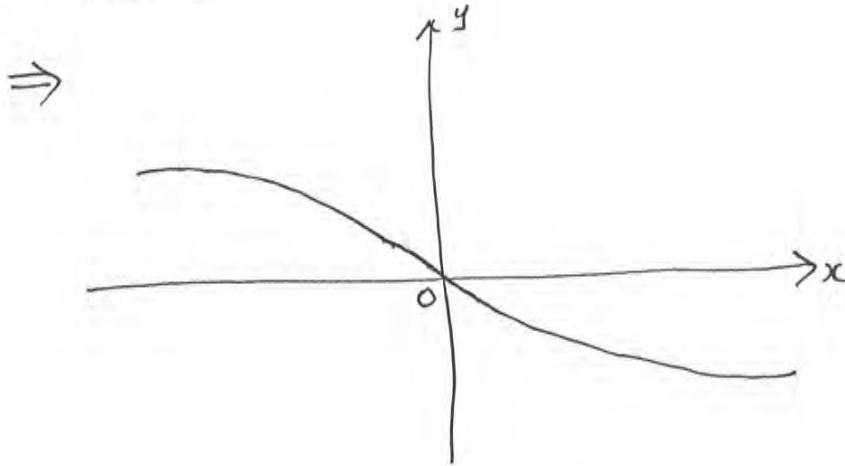
$$(iii) \quad f'(x) = \frac{1}{\sqrt{x^2+1}-x} \cdot \frac{1}{2}(x^2+1)^{-\frac{1}{2}} \cdot 2x - 1$$

$$= \frac{x - \sqrt{x^2+1}}{\sqrt{x^2+1}(\sqrt{x^2+1}-x)} = \frac{-1}{\sqrt{x^2+1}}$$

(iv) For $x > 0$, $f''(x) > 0$ i.e. concave upward

& for $x < 0$, $f''(x) < 0$ i.e. concave downward

$$f'(0) = -1$$



(v) Put $u = \ln(\sqrt{x^2+1} - x)$, $\frac{dv}{dx} = 1$

$$\Rightarrow \frac{du}{dx} = -\frac{1}{\sqrt{x^2+1}}, \quad v = x$$

$$\therefore I = \left[x \ln(\sqrt{x^2+1} - x) \right]_0^1 + \int_0^1 \frac{x}{\sqrt{x^2+1}} dx$$

For $\int_0^1 \frac{x}{\sqrt{x^2+1}} dx$, put $u = x^2+1$; $x=0, u=1$
 $\frac{du}{dx} = 2x$ $x=1, u=2$

$$\therefore I = \ln(\sqrt{2}-1) + \frac{1}{2} \int_1^2 \frac{1}{\sqrt{u}} du$$

$$= \ln(\sqrt{2}-1) + \frac{1}{2} \cdot 2 [\sqrt{u}]_1^2$$

$$= \ln(\sqrt{2}-1) + \sqrt{2} - 1$$

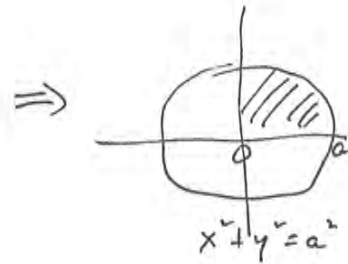
Question 4

$$(a) A = 4 \int_0^a b \sqrt{1 - \frac{x^2}{a^2}} dx$$

$$= \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx$$

$$= \frac{4b}{a} \cdot \frac{1}{4} \pi a^2$$

$$= \pi a b$$



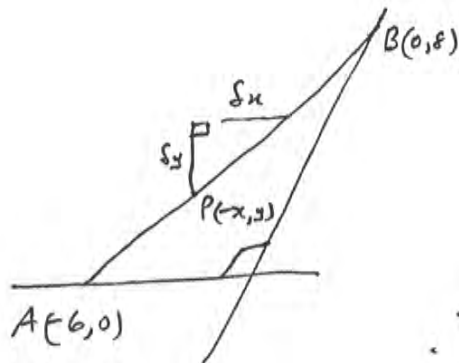
(b) Major axis length = x , minor axis length = $\frac{x}{2}$

$$\Rightarrow 2a = x, 2b = \frac{x}{2}$$

$$\text{or } a = \frac{x}{2}, b = \frac{x}{4}$$

$\therefore$ Area of slice = $\frac{1}{2} \cdot \pi \cdot \frac{x^2}{8}$ from (a)

$$= \frac{\pi x^2}{16}$$



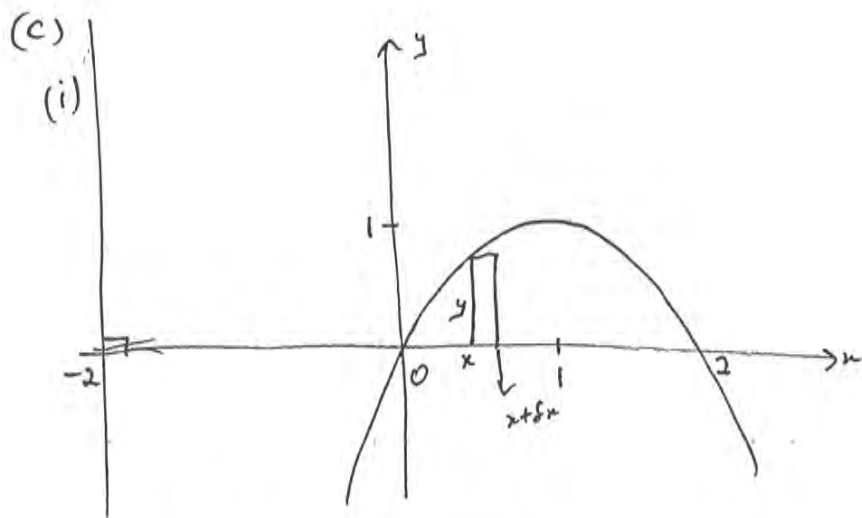
$$\Rightarrow \delta V \approx \frac{\pi x^2}{16} \delta y \quad \text{where } \frac{\delta y}{\delta x} = \frac{8}{6} = \frac{4}{3}$$

$$\sim \delta y = \frac{4}{3} \delta x$$

$$\therefore V = \int_0^6 \frac{\pi x^2}{16} \frac{4}{3} dx$$

$$= \frac{\pi}{12} \left[\frac{x^3}{3} \right]_0^6$$

$$= 6\pi$$



(i) at $x=1$, $y=1$

$\therefore$ using Simpson's rule (exact here, of course),

$$A = \frac{1}{6} \cdot 2 [0 + 0 + 4] = \frac{4}{3}$$

(ii) $\delta V \approx \pi \left((x+\delta x + 2)^2 - (x+2)^2 \right) y$

$$\approx \pi \cdot 2(x+2) \delta x y$$

$$\Rightarrow V = 2\pi \int_0^2 (x+2)x(2-x) dx$$

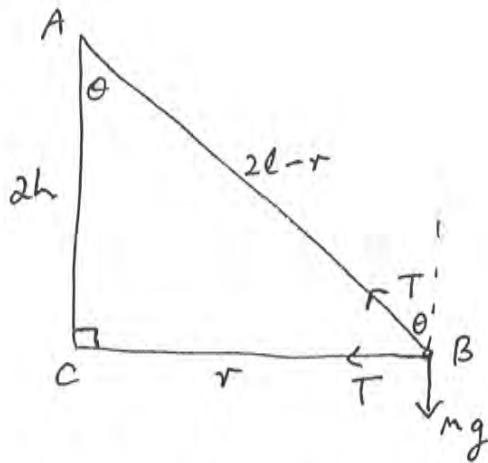
$$= 2\pi \int_0^2 (4x - x^2) dx$$

$$= 2\pi \left[2x^2 - \frac{x^3}{3} \right]_0^2$$

$$= 2\pi (8 - \frac{8}{3}) = \frac{16\pi}{3}$$

Question 5

(a)



(i) $AB > BC \Rightarrow 2l - r > r$ i.e. $r < l$

(ii) Resolving vertically at B, $mg = T \cos \theta$
i.e. $T = mg \sec \theta$

(iii) Resolving in direction $\vec{BC}$,
 $m\omega^2 r = T + T \sin \theta$
 $= mg \sec \theta (1 + \sin \theta)$
 $\Rightarrow \omega^2 r = g (\sec \theta + \tan \theta)$

(iv) $\sec \theta + \tan \theta = \frac{2l - r}{2h} + \frac{r}{2h} = \frac{2l}{2h} = \frac{l}{h}$

$$\therefore \omega^2 r = g \frac{l}{h} \Rightarrow r = \frac{gl}{h\omega^2}$$

(v) Since $r < l$, $\frac{gl}{h\omega^2} < l$

$$\text{or } \omega^2 > \frac{g}{h}$$

$$\text{i.e. } \omega > \sqrt{\frac{g}{h}}$$

(b) (i) For $n=1$, $LS = \frac{4 \cdot 1}{1 \cdot 3} - \frac{4 \cdot 2}{3 \cdot 5} = \frac{4}{5}$

$$RS = 1 - \frac{1}{5} = \frac{4}{5}$$

$\therefore$ Assume $\frac{4 \cdot 1}{1 \cdot 3} - \dots + \frac{(-1)^n 4(n+1)}{(2n+1)(2n+3)} = 1 + (-1)^n \frac{1}{2n+3}$ for some positive integer n

$$\begin{aligned} \text{Then } & \frac{4 \cdot 1}{1 \cdot 3} - \dots + \frac{(-1)^n 4(n+1)}{(2n+1)(2n+3)} + \frac{(-1)^{n+1} 4(n+2)}{(2n+3)(2n+5)} \\ &= 1 + (-1)^n \frac{1}{2n+3} + \frac{(-1)^{n+1} 4(n+2)}{(2n+3)(2n+5)}, \text{ using the assumption} \\ &= 1 + \frac{(-1)^{n+1}}{2n+3} \left(\frac{4(n+2)}{2n+5} - 1 \right) \\ &= 1 + \frac{(-1)^{n+1}}{2n+3} \left(\frac{4n+8-2n-5}{2n+5} \right) \\ &= 1 + \frac{(-1)^{n+1}}{2n+3} \frac{(2n+3)}{2n+5} \\ &= 1 + (-1)^{n+1} \cdot \frac{1}{2n+5} \end{aligned}$$

$\therefore$ by induction it's true for $n \geq 1$

(ii) $S = \lim_{n \rightarrow \infty} \left(1 + \frac{(-1)^n}{2n+3} \right) = 1$

(c) $z^2 + 1 = \cos 2\theta + i \sin 2\theta + 1$
 $= 2\cos^2 \theta + 2i \sin \theta \cos \theta$
 $= 2\cos \theta (\cos \theta + i \sin \theta)$

$$\Rightarrow |z^2 + 1| = 2\cos \theta, \arg(z^2 + 1) = \theta$$

Question 6

(a) (i) l_1 is $\frac{x}{a} - \frac{y}{b} = 0$

$\therefore$ at A , $\frac{ae}{a} - \frac{y}{b} = 0 \Rightarrow y = be$

$\therefore A = (ae, be)$

(ii) $\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = \frac{b^2 x}{a^2 y} = \frac{b \sec \theta}{a \tan \theta}$ at P

$\therefore$ Tangent at P is $y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$

or $\frac{\tan \theta}{b} y - \tan^2 \theta = \frac{\sec \theta}{a} x - \sec^2 \theta$

i.e. $\frac{\sec \theta}{a} x - \frac{\tan \theta}{b} y = \sec^2 \theta - \tan^2 \theta$

(iii) From (ii), at A , $x = ae$, $y = be$

$\Rightarrow e \sec \theta - e \tan \theta = \sec^2 \theta - \tan^2 \theta$

$\Rightarrow e = \frac{(\sec^2 \theta - \tan^2 \theta)}{\sec \theta - \tan \theta} (\sec \theta + \tan \theta) = \sec \theta + \tan \theta$

(iv) l_2 is $\frac{x}{a} + \frac{y}{b} = 0$ with gradient $-\frac{b}{a}$

$\text{grad } SP = \frac{b \tan \theta}{a \sec \theta - ae} = -\frac{b}{a} \frac{\tan \theta}{\tan \theta} \quad \text{from (iii)}$
 $= -\frac{b}{a}$

$\therefore l_2 \parallel SP$

$$(b) (i) \frac{dv}{dt} = \frac{16 + v^2}{v}$$

$$\Rightarrow \frac{dt}{dv} = \frac{v}{16 + v^2}$$

$$\Rightarrow [t]_0^T = \frac{1}{2} \left[\ln(16 + v^2) \right]_V^{4V}$$

$$\therefore T = \frac{1}{2} \left(\ln(16 + 16V^2) - \ln(16 + V^2) \right)$$

$$= \frac{1}{2} \left(\ln 16 + \ln\left(\frac{1+V^2}{16+V^2}\right) \right)$$

$$= \frac{1}{2} \left(4 \ln 2 + \ln\left(\frac{1+V^2}{16+V^2}\right) \right)$$

$$= 2 \ln 2 + \frac{1}{2} \ln\left(\frac{1+V^2}{16+V^2}\right)$$

$$< 2 \ln 2 \quad \text{since} \quad \frac{1+V^2}{16+V^2} < 1$$

$$\Rightarrow \ln\left(\frac{1+V^2}{16+V^2}\right) < 0$$

$$(ii) \quad v \frac{dv}{dx} = \frac{16 + v^2}{v}$$

$$\Rightarrow \frac{dx}{dv} = \frac{v^2}{16 + v^2} = 1 - \frac{16}{16 + v^2}$$

$$\therefore [x]_0^X = \left[v - \frac{16}{4} \tan^{-1} \frac{v}{4} \right]_V^{4V}$$

$$\Rightarrow X = 4V - 4 \tan^{-1} V - \left(V - 4 \tan^{-1} \frac{V}{4} \right)$$

$$= 3V - 4 \left(\tan^{-1} V - \tan^{-1} \frac{V}{4} \right)$$

$$= 3V - 4 \tan^{-1} \left(\frac{V - \frac{V}{4}}{1 + V \cdot \frac{V}{4}} \right) = 3V - 4 \tan^{-1} \left(\frac{3V}{4 + V^2} \right)$$

Question 7

(a) (i) at R, $y=0 \therefore \frac{x}{a} \cos\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha-\beta}{2}\right)$

$$\therefore x = a \frac{\left(\cos \frac{\alpha}{2} \cos \frac{\beta}{2} + \sin \frac{\alpha}{2} \sin \frac{\beta}{2}\right)}{\cos \frac{\alpha}{2} \cos \frac{\beta}{2} - \sin \frac{\alpha}{2} \sin \frac{\beta}{2}}$$

$$\text{i.e. } x = a \frac{1 + \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}$$

(ii) From (i), $\frac{ae}{a} \cos\left(\frac{\alpha+\theta}{2}\right) = \cos\left(\frac{\alpha-\theta}{2}\right)$

$$\text{i.e. } e \cos\left(\frac{\alpha+\theta}{2}\right) = \cos\left(\frac{\alpha-\theta}{2}\right)$$

(iii) Now $e \left(\cos \frac{\alpha}{2} \cos \frac{\theta}{2} - \sin \frac{\alpha}{2} \sin \frac{\theta}{2}\right) = \cos \frac{\alpha}{2} \cos \frac{\theta}{2} + \sin \frac{\alpha}{2} \sin \frac{\theta}{2}$

$$\therefore \cos \frac{\alpha}{2} \cos \frac{\theta}{2} (e-1) = \sin \frac{\alpha}{2} \sin \frac{\theta}{2} (e+1)$$

$$\Rightarrow \tan \frac{\alpha}{2} \tan \frac{\theta}{2} = \frac{e-1}{e+1}$$

(iv) From (iii), using chord $\alpha\alpha'$,

$$\tan \frac{\beta}{2} \tan \frac{\pi+\theta}{2} = \frac{e-1}{e+1}$$

$$\text{or } \tan \frac{\beta}{2} \tan \left(\frac{\pi}{2} + \frac{\theta}{2}\right) = \frac{e-1}{e+1}$$

$$\text{or } -\tan \frac{\beta}{2} \tan \left(\frac{\pi}{2} - \frac{\theta}{2}\right) = \frac{e-1}{e+1} = -\tan \frac{\beta}{2} \cot \frac{\theta}{2}$$

(v) From (iii) and (iv),

$$-\tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \left(\frac{e-1}{e+1} \right)^2 \text{ is constant}$$

$$\text{But, at } R, \quad x = \frac{a \left(1 + \tan \frac{\alpha}{2} \tan \frac{\beta}{2} \right)}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}} \quad \text{from (i)}$$

$\Rightarrow$ at R , x is constant

i.e. R is a fixed point

$$(b) (i) \quad LS = (1-x^b)^{n-1} (1 - (1-x^b)) = x^b (1-x^b)^{n-1}$$

$$(ii) \quad u_0 = \int_0^1 x^{a-1} dx = \left[\frac{x^a}{a} \right]_0^1 = \frac{1}{a}$$

$$(iii) \quad \text{Put } u = (1-x^b)^n, \quad \frac{du}{dx} = -bx^{b-1}(1-x^b)^{n-1}$$

$$\therefore \frac{du}{dx} = -nbx^{b-1}(1-x^b)^{n-1}, \quad v = \frac{x^a}{a}$$

$$\therefore u_n = \left[\frac{x^a}{a} (1-x^b)^n \right]_0^1 + \frac{nb}{a} \int_0^1 x^a \cdot x^{b-1} (1-x^b)^{n-1} dx$$

$$\Rightarrow u_n = \frac{nb}{a} \int_0^1 x^{a-1} x^b (1-x^b)^{n-1} dx$$

$$= \frac{nb}{a} \int_0^1 x^{a-1} ((1-x^b)^{n-1} - (1-x^b)^n) dx \quad \text{from (i)}$$

$$\text{i.e. } u_n = \frac{nb}{a} (u_{n-1} - u_n)$$

$$\text{or } au_n = nbu_{n-1} - nbu_n$$

$$\Rightarrow (a+bn)u_n = nbu_{n-1}$$

Question 8

$$(a) (i) \quad \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2\alpha\beta \\ = 0^2 - 2\left(-\frac{A}{2}\right) = A$$

$$(ii) \quad \frac{\beta}{\gamma} + \frac{\gamma}{\beta} = \frac{\beta^2 + \gamma^2}{\beta\gamma} = \frac{(A - \alpha^2)\alpha}{\alpha\beta\gamma} \\ = A\alpha - \alpha^3 \quad \text{since } \alpha\beta\gamma = \frac{1}{2} = 1$$

$$(iii) \quad \text{From (ii), put } x = A\alpha - \alpha^3$$

$$\text{Now, } P(\alpha) = 2\alpha^3 - A\alpha - 2 = 0$$

$$\therefore x = \alpha^3 - 2 \quad \text{or } \alpha^3 = x + 2$$

$$\therefore \text{cubic is } 2(x+2) - A(x+2)^{\frac{1}{3}} - 2 = 0$$

$$\text{or } A(x+2)^{\frac{1}{3}} = 2(x+1)$$

$$\Rightarrow A^3(x+2) = 8(x+1)^3$$

$$\text{i.e. } 8(x+1)^3 - A^3(x+2) = 0$$

$$(b) (i) \quad S = \frac{(1+x)^n - 1}{1+x-1} = \frac{(1+x)^n - 1}{x}, \quad x \neq 0$$

$$(ii) \quad S = \frac{1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n - 1}{x}$$

$$= \binom{n}{1} + \binom{n}{2}x + \dots + \binom{n}{n}x^{n-1}$$

(iii) When $x=0$

$$\binom{n}{1} + \binom{n}{2}x + \dots + \binom{n}{n}x^{n-1} = \binom{n}{1} = n$$

$$1 + (1+x) + (1+x)^2 + \dots + (1+x)^{n-1} = n \times 1 = n$$

$\therefore$ True for $n=0$.

$$(iv) \text{ Now } S = \binom{n}{1} + \binom{n}{2}x + \dots + \binom{n}{n}x^{n-1} = 1 + (1+x) + \dots + (1+x)^{n-1}$$

$$\therefore \int_{-1}^0 \left(\binom{n}{1} + \binom{n}{2}x + \dots + \binom{n}{n}x^{n-1} \right) dx = \int_{-1}^0 \left(1 + (1+x) + \dots + (1+x)^{n-1} \right) dx$$

$$\Rightarrow \left[\binom{n}{1}x + \binom{n}{2}\frac{x^2}{2} + \dots + \binom{n}{n}\frac{x^n}{n} \right]_{-1}^0 = \left[x + \frac{(1+x)^2}{2} + \dots + \frac{(1+x)^n}{n} \right]_{-1}^0$$

$$\therefore - \left(-\binom{n}{1} + \binom{n}{2}\frac{1}{2} - \dots + \binom{n}{n}\frac{(-1)^n}{n} \right) = \left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) - (-1)$$

$$\text{i.e. } \binom{n}{1} - \binom{n}{2}\frac{1}{2} + \dots + (-1)^{n+1}\binom{n}{n}\frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$